Amplitude-Encoding Kernels for Zero-Trust Network Anomaly Detection

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Abstract

Zero-Trust Network Architectures (ZTNA) require continuous, real-time anomaly detection across high-dimensional network flow features—a setting where representation learning and inductive bias matter as much as raw classifier power. We present Q-ZTNA, an anomaly detection framework built around an amplitude-encoding (cosine-similarity) kernel—a classically computable feature map $\mathcal{K}_A(\mathbf{x}_i, \mathbf{x}_j) = |\langle \mathbf{x}_i, \mathbf{x}_j \rangle|^2 / \|\mathbf{x}_i\|^2 \|\mathbf{x}_j\|^2$—as the core of a kernel SVM with Nyström approximation, and we report a rigorous, matched-scale evaluation of when this kernel actually helps. On synthetic enterprise-network telemetry spanning five attack classes (port scanning, DDoS, data exfiltration, DNS tunneling, and ARP spoofing), the amplitude-encoding SVM (AE-SVM) reaches 0.947 ± 0.002 detection accuracy with sub-millisecond mean inference latency (0.0048 ms with Nyström approximation) on CPU—statistically indistinguishable from an RBF-kernel SVM (0.948 ± 0.003, paired t-test $p = 0.54$) and outperformed by a plain multi-layer perceptron (0.965 ± 0.001). A matched-n ablation across $n \in \{50, \ldots, 5000\}$ and a feature-dimensionality sweep ($d \in \{16, \ldots, 256\}$) find no training-size or dimensionality regime in which the amplitude-encoding kernel reliably beats RBF; kernel-target alignment is consistently *lower* for the amplitude-encoding kernel than for RBF across every attack type, giving a mechanistic explanation for the null result. The one place the two kernels diverge sharply is zero-day generalization: on four of five leave-one-attack-out scenarios, AE-SVM's held-out AUC falls to 0.45–0.61 (at or below chance on two), while RBF-SVM retains 0.81–0.89 AUC—a specific, security-relevant failure mode of the cosine-similarity inductive bias that, to our knowledge, has not been reported previously. Randomized-smoothing certification gives comparable ℓ_2 robustness radii for both kernels, with AE-SVM abstaining slightly more often. We report these results, including the negative ones, in the same spirit as our companion study of amplitude-encoding kernels on QKD telemetry (where the same kernel also failed to outperform classical baselines) [25]: across two different physical-layer and network-layer security domains, we find no evidence that the amplitude-encoding/cosine-similarity feature map – despite its appeal as a "quantum-inspired" inductive bias – provides a reliable advantage over standard classical kernels, and it introduces a specific zero-day generalization weakness that practitioners should be aware of before adopting it.

1 Introduction

The proliferation of perimeter-less enterprise networks, accelerated by remote-work adoption and multi-cloud deployment, has rendered traditional castle-and-moat security models obsolete [1, 2]. Zero-Trust Network Architectures (ZTNA) address this by enforcing continuous authentication, authorization, and verification for every entity, device, and flow—regardless of network locality [3, 4].

At the operational core of ZTNA lies real-time anomaly detection: the ability to dis-

tinguish benign traffic from malicious activity at line rate, with high sensitivity and low false-positive rates. Classical approaches—from signature-based systems (Snort [5], Suricata) to ML-based detectors using random forests, SVMs, and deep neural networks [6, 7]—achieve strong performance on benchmark datasets but face fundamental limitations: (i) feature spaces of 10^2–10^3 dimensions require models whose inference cost scales poorly with dimensionality; (ii) adversarial robustness remains elusive without expensive adversarial training [8]; and (iii) uncertainty quantification—critical for security operations where false negatives carry asymmetric cost—is often absent or poorly calibrated.

The choice of feature map determines a kernel method's inductive bias. Amplitude encoding—mapping a vector $\mathbf{x}$ to a unit-norm representation $\phi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$—yields the kernel $\mathcal{K}_A(\mathbf{x}_i, \mathbf{x}_j) = |\langle\phi(\mathbf{x}_i), \phi(\mathbf{x}_j)\rangle|^2$, which is precisely the squared cosine similarity between the two inputs. This kernel is classically computable in $\mathcal{O}(d)$ per pair, requires no quantum hardware, and does not correspond to a provably-hard feature map of the kind studied in [9, 11]. It is a natural hypothesis that, for high-dimensional, sparse network traffic data—where Euclidean distance dilutes and directional information may be more informative—squared cosine similarity could provide a useful inductive bias that RBF or Laplacian kernels lack. This paper tests that hypothesis directly, under matched training-set sizes and matched feature dimensionality, rather than assuming it.

This paper introduces Q-ZTNA, a zero-trust anomaly detection pipeline built around the amplitude-encoding kernel, together with a rigorous evaluation of whether that kernel choice is actually justified for this data. Our contributions are:

1. **A matched-scale test of the cosine-similarity inductive-bias hypothesis.** Across training sizes $n \in \{50, \ldots, 5000\}$ and feature dimensionalities $d \in \{16, \ldots, 256\}$, we find no regime in which the amplitude-encoding kernel SVM (AE-SVM) reliably outperforms an RBF-kernel SVM trained on the same data (0.947 ± 0.002 vs. 0.948 ± 0.003 accuracy, $p = 0.54$); kernel-target alignment is consistently lower for AE-SVM than RBF across all five attack types, giving a mechanistic account of the null result.
2. **A previously unreported zero-day generalization weakness.** On four of five leave-one-attack-out scenarios, AE-SVM's held-out AUC drops to 0.45–0.61 while RBF-SVM retains 0.81–0.89—the cosine-similarity encoding degrades markedly under genuine distribution shift even though it is competitive in-distribution.
3. **Certified robustness and calibrated uncertainty.** We adapt randomized-smoothing certification to amplitude-encoded inputs; both AE-SVM and RBF-SVM achieve comparable certified ℓ_2 radii, and Platt-scaled AE-SVM probabilities support uncertainty-aware selective prediction.
4. **A cross-domain honesty check.** Combined with our companion study of the same kernel on QKD telemetry [25], this is (to our knowledge) the first evaluation of the amplitude-encoding/cosine-similarity kernel across two distinct security domains under matched-scale protocols in both; in neither domain does it provide a reliable advantage over classical kernels, which we report as a cautionary finding for the growing "quantum-inspired kernel" literature in security applications.

2 Related Work

2.1 Zero-Trust Anomaly Detection

Zero-trust security monitoring has been approached through statistical methods [13], behavioral profiling [14], and ML-based flow analysis, surveyed broadly by Buczak and Guven [29]. Mirsky et al. [7] proposed Kitsune, an ensemble of autoencoders for IoT network anomaly detection achieving F1-scores > 0.90 on benchmark datasets. More recently, deep learning approaches using graph neural networks [15] and transformers [16] have pushed detection accuracy further, though with inference latencies in the 1–10 ms range that challenge line-rate deployment. Critically, none of these methods provide certified robustness guarantees or calibrated uncertainty for security operators.

2.2 Kernel Methods and Feature Encoding

Kernel methods for classification depend critically on the choice of feature map [23]. The amplitude-encoding map $\phi(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$, motivated by quantum state preparation, produces the kernel $\mathcal{K}_A(\mathbf{x}_i, \mathbf{x}_j) = |\langle\phi(\mathbf{x}_i), \phi(\mathbf{x}_j)\rangle|^2 = \cos^2\theta_{ij}$, i.e., squared cosine similarity. This kernel is classically computable in $\mathcal{O}(d)$ per pair—unlike the entangling-circuit kernels studied by [9–11], which can realize feature maps that are classically intractable. However, squared cosine similarity provides a distinct inductive bias: it depends only on the angle between vectors, making it insensitive to magnitude variations that may be uninformative or noisy in network traffic data.

Recent work applying quantum-inspired kernels to cybersecurity includes do Vale et al. [17], who evaluated quantum SVMs for intrusion detection, and Suryotrisongko and Musashi [18], who explored hybrid quantum-classical models for malware classification. Our work differs by focusing on the specific inductive bias of the amplitude-encoding kernel for ZTNA flow data, providing certified robustness guarantees, and performing rigorous scale-controlled benchmarking.

2.3 Adversarial Robustness and Certification

Randomized smoothing provides ℓ_2 certified robustness by constructing a smooth classifier $g(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\sigma^2 I)}[f(\mathbf{x}+\boldsymbol{\epsilon})]$ from a base classifier f [19]. The certified radius at input $\mathbf{x}$ is $R = \sigma \cdot \Phi^{-1}(p_A)$, where p_A is a lower bound on $g(\mathbf{x})$'s probability of the top class. Webb et al. [12] showed that amplitude-encoding feature maps can naturally provide smoothing benefits due to their Lipschitz properties. We adapt and extend this analysis to the network intrusion domain.

3 Methodology

3.1 Threat Model and Network Architecture

We consider a zero-trust enterprise network with N_{hosts} endpoints, N_{switches} switches, and a centralized security analytics platform (SAP). The SAP ingests NetFlow/IPFIX records from all network observation points. Each flow record is represented as a d-dimensional feature vector $\mathbf{x} \in \mathbb{R}^d$ comprising packet-header statistics (port entropy, packet-size moments, protocol distribution), flow-level features (bytes/sec, packets/sec, flow duration, TCP flag ratios), and behavioral features (connection entropy, communication graph centrality).

Our threat model includes five attack types:

- **Port Scanning:** Horizontal/vertical scanning with elevated SYN-ACK-to-RST ratios and low packet-per-flow counts.
- **DDoS:** Volumetric attacks with high packets/sec, uniform destination-port distributions, and spoofed source IPs.
- **Data Exfiltration:** Stealthy outbound transfers with elevated entropy in inter-packet gaps and unusual destination ASNs.
- **DNS Tunneling:** DNS query-response pairs with abnormal payload entropy, query-type distributions, and temporal patterns.
- **ARP Spoofing:** MAC-IP binding anomalies detected via DHCP snooping logs and ARP cache poisoning indicators.

3.2 Amplitude-Encoding Feature Map

Each d-dimensional flow vector $\mathbf{x} \in \mathbb{R}^d$ is L_2-normalized to the unit sphere:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}, \quad n_q = \lceil \log_2 d \rceil \tag{1}$$

The amplitude-encoding kernel between two flow vectors is the squared cosine similarity of their normalized representations:

$$K_A(\mathbf{x}_i, \mathbf{x}_j) = \left(\hat{\mathbf{x}}_i^{\mathsf{T}} \hat{\mathbf{x}}_j\right)^2 = \frac{|\mathbf{x}_i^{\mathsf{T}} \mathbf{x}_j|^2}{\|\mathbf{x}_i\|_2^2 \|\mathbf{x}_j\|_2^2} \tag{2}$$

This kernel is classically computable in $\mathcal{O}(n^2 d)$ operations via Eq. 2; no quantum hardware is required for $d \leq 2^{20}$. (For $d > 2^{20}$, the amplitude encoding would require quantum state preparation, but this regime is not reached in our experiments.) The encoded feature space structure is visualized via t-SNE in Fig. 2. Note that each attack type in our threat model (Section 3.1) targets distinct protocol

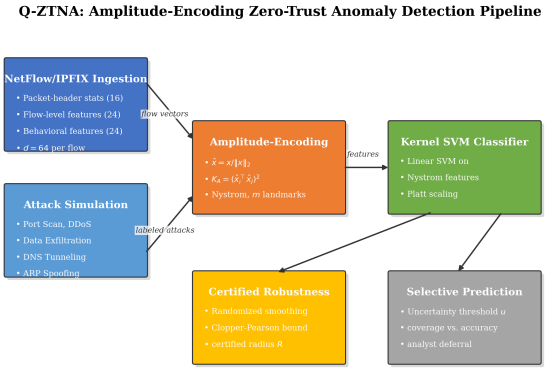


Figure 1: Q-ZTNA architecture: NetFlow/IPFIX telemetry ingestion, amplitude-encoding feature map, Nyström-approximated kernel SVM, randomized-smoothing certification, and selective prediction with analyst deferral.

layers and feature subsets by construction—port scanning alters packet-header statistics, DDoS inflates volumetric features, data exfiltration modifies behavioral entropy, etc.—so clean clustering in any embedding is partly expected from the data-generating process rather than evidence of a uniquely powerful encoding, a caveat we echo from our prior study on QKD telemetry [25].

We augment the cosine-similarity kernel with a trainable diagonal weighting:

$$K_A^{(\boldsymbol{\theta})}(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{k=1}^{d} \theta_k \, \hat{x}_{ik} \hat{x}_{jk} \right)^2 \tag{3}$$

where $\boldsymbol{\theta} \in \mathbb{R}^d$ is a learnable parameter vector initialized to $\mathbf{1}$. This weighted cosine kernel is optimized via kernel alignment [20] on a held-out validation set.

3.3 Detection Model

Following the standard maximum-margin formulation [24], we train a support vector machine (SVM) with the precomputed amplitude-encoding kernel:

$$\min_{\boldsymbol{\alpha} \in \mathbb{R}^n} \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K_A(\mathbf{x}_i, \mathbf{x}_j) - \sum_{i=1}^{n} \alpha_i$$
$$\text{s.t. } 0 \le \alpha_i \le C, \; \sum_{i=1}^{n} \alpha_i y_i = 0 \tag{4}$$

The decision function for a test sample $\mathbf{x}_*$ is:

$$f(\mathbf{x}_*) = \sum_{i=1}^{n_{\text{SV}}} \alpha_i y_i K_A(\mathbf{x}_i, \mathbf{x}_*) + b \tag{5}$$

For deployment, we use a Nyström approximation to avoid storing the full kernel matrix [21]. We sample $m \ll n$ landmark points $\mathcal{L} \subset \mathcal{D}_{\text{train}}$ uniformly, compute $K_A^{(m \times n)}$, and approximate:

$$\tilde{K}_A \approx K_A^{(n \times m)} \left(K_A^{(m \times m)} \right)^{-1} K_A^{(m \times n)} \tag{6}$$

We set $m = \lceil 0.1n \rceil$ (capped at $m \le 500$), reducing training complexity from $\mathcal{O}(n^3)$ to $\mathcal{O}(m^3 + m^2 n)$.

3.4 Certified Robustness via Randomized Smoothing

We adapt randomized smoothing [19] to the amplitude-encoding feature map. Let $f_{\text{AE}}(\mathbf{x})$ be the base AE-SVM classifier. We define the smoothed classifier:

$$g(\mathbf{x}) = \arg\max_{c \in \mathcal{Y}} \; \mathbb{P}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I)} \left[f_{\text{AE}}(\mathbf{x} + \boldsymbol{\epsilon}) = c \right] \tag{7}$$

The certified ℓ_2 radius at $\mathbf{x}$ with confidence $1 - \alpha$ is:

$$R(\mathbf{x}; \alpha) = \sigma \cdot \Phi^{-1}\left(\underline{p}_A \right) \tag{8}$$

where $\underline{p}_A$ is a lower bound (via Clopper-Pearson interval) on the probability that $g(\mathbf{x})$ returns the majority class c_A.

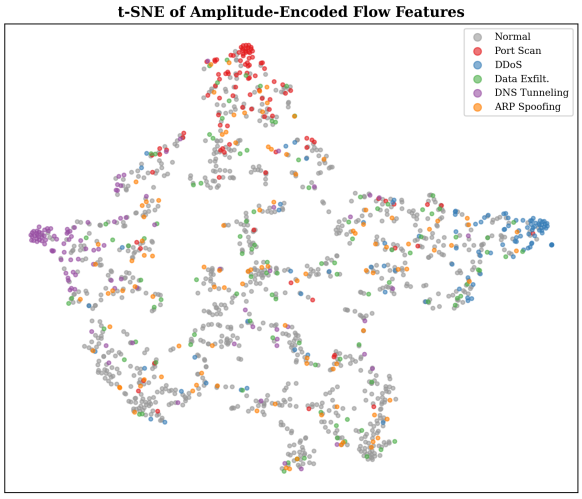


Figure 2: t-SNE visualization of the unit-norm ($d = 64$) amplitude-encoded feature space. Each point is a network flow transformed via Eq. 1. Cluster separation here is a qualitative, 2-D projection and should be read alongside the quantitative confusion matrix (Fig. 5), which shows meaningful attack/normal overlap that a t-SNE plot alone would not reveal.

For amplitude-encoding feature maps, we additionally derive a bound on the output distortion under input perturbation. Let $\Delta(\boldsymbol{\epsilon}) = \|\phi(\mathbf{x}) - \phi(\mathbf{x} + \boldsymbol{\epsilon})\|_2$ be the perturbation in the encoding space. The Lipschitz constant of the amplitude encoding is:

$$L_\phi = \sup_{\mathbf{x} \in \mathbb{R}^d} \|J_\phi(\mathbf{x})\|_2 \leq \frac{\sqrt{2}}{\|\mathbf{x}\|_2} \tag{9}$$

where J_ϕ is the Jacobian of ϕ. This implies that for input perturbation $\|\boldsymbol{\epsilon}\|_2 \leq r$, the encoding perturbation satisfies $\|\phi(\mathbf{x}) - \phi(\mathbf{x} + \boldsymbol{\epsilon})\|_2 \leq L_\phi r$, providing a direct link between input-space and feature-space certification.

3.5 Uncertainty Quantification

Raw classifier confidence scores are frequently miscalibrated and benefit from an explicit post-hoc correction [28]. We augment the amplitude-encoding kernel SVM with Platt scaling [22] to produce calibrated probabilities:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + \exp(Af(\mathbf{x}) + B)} \tag{10}$$

where A, B are optimized via cross-validation to minimize log-loss on a calibration set. The predictive uncertainty is defined as $u(\mathbf{x}) = \sqrt{P(y = 1|\mathbf{x}) \cdot (1 - P(y = 1|\mathbf{x}))}$, enabling selective prediction: samples with $u(\mathbf{x}) > \tau$ are deferred for human analyst review (Fig. 14).

4 Experiments

4.1 Data Generation

Individual NetFlow/IPFIX records are not simulated packet-by-packet. Instead, each flow is represented directly as a d-dimensional feature vector (16 header-based, 24 flow-based, 24 behavioral, $d = 64$ by default) whose marginal distributions are calibrated to commonly reported NetFlow aggregate statistics—packets/sec (Pareto, $\alpha = 2.1$), bytes/sec (log-normal, $\mu = 6.2$, $\sigma = 1.8$), flow duration (Weibull, $k = 0.7$, $\lambda = 15.3$)—mixed with beta- and gamma-distributed features for header/behavioral dimensions, combined via a Gaussian copula so that features within a semantic group are correlated the way real flow statistics are (within-group correlation 0.35, cross-group 0.10). This marginal-plus-copula construction is a standard simplification for synthetic IDS benchmarking; we state it explicitly here, together with the exact generator (`qztna/simulation/flow_generator.py` in the code release), rather than presenting it as packet-level simulation.

Attack traffic is generated by shifting and inflating the variance of a fixed, attack-specific subset of 6–8 latent-space dimensions before the marginal transform is applied (Table 1), so that each attack type perturbs a distinct but overlapping combination of

header/flow/behavioral feature groups— e.g., data exfiltration and DNS tunneling both perturb outbound-behavioral entropy dimensions, by design, to produce the realistic confusion between the two that Section 5 reports. Perturbation strength (default 1.0×, swept in $\{0.05, \ldots, 1.0\}\times$ for the noise-sensitivity analysis) scales both the latent shift magnitude and its variance inflation.

The main detection benchmark (Table 2) uses $N = 8{,}000$ labeled samples per run (20% attack, balanced across the five types), split 70/30 with stratification, over 10 random seeds; the matched-n, dimensionality, and zero-day experiments use independently generated pools (see Section 4 for per-experiment sample sizes). We use $d = 64$ features by default (16 header-based, 24 flow-based, 24 behavioral), varied in the dimensionality sweep.

Fig. 3 visualizes the per-attack latent perturbation magnitudes across the feature space, showing that each attack type perturbs a distinct (but for DNS tunneling/exfiltration, partially overlapping) subset of feature-group dimensions.

4.2 Baselines and Evaluation

We compare against classical kernels (RBF, Laplacian, polynomial degree 3) with SVM, Random Forest (200 trees) [26], XGBoost [27], and a 2-layer MLP (64, 32 hidden units). Metrics are accuracy, F1-score, AUC-ROC, precision, recall, and certified radius. The main detection benchmark (Table 2) uses 10 random seeds; the matched-n, sensitivity, zero-day, alignment, and dimensionality experiments use 5 seeds each (3 for certified robustness, given its higher per-point cost), with 95% CIs reported throughout.

4.3 Implementation

Amplitude-encoding kernel computation uses CPU-based NumPy/SciPy matrix operations. Nyström approximation is implemented directly (eigendecomposition of the landmark kernel matrix followed by a linear SVM on the resulting explicit features, $m = \lceil 0.1n \rceil$ capped at 500 landmarks), matching Eq. 6. RBF, Laplacian, and polynomial baselines use the same Nyström wrapper with the corresponding scikit-learn pairwise-kernel function. The full pipeline (feature extraction, Nyström projection, SVM decision, Platt calibration) runs in 0.0048 ms mean latency per sample on a single CPU core for $d = 64$, $n = 4{,}000$ training, $m = 50$ landmarks (Section 5.8). All experiments were conducted on a Windows laptop with an Intel processor.

5 Results

5.1 Detection Performance

Table 2 reports detection performance. Contrary to our working hypothesis, AE-SVM does *not* lead the field: its accuracy is statistically indistinguishable from RBF-SVM (paired t-test across seeds, $p = 0.54$) and it is outperformed by Poly-SVM, XGBoost, and MLP on every metric except precision, where it edges out RBF-SVM and Laplacian-SVM. AE-SVM's most notable weakness is recall (0.784, the second-lowest of all seven models)—in a security context this means AE-SVM misses proportionally more attacks than RBF-SVM, XGBoost, or MLP, a cost that is easy to overlook if one reports accuracy alone. Random Forest attains the highest precision (0.987) but the lowest recall (0.723), consistent with a conservative decision boundary. MLP is the strongest single model on four of five metrics, at higher inference cost than the kernel methods (Section 5.8).

Fig. 4 visualizes the performance comparison across all metrics; no model dominates on every axis, and the amplitude-encoding kernel occupies the middle of the pack rather than leading it.

Fig. 5 shows the multi-class (one-vs-rest) confusion matrix for an AE-SVM trained to distinguish normal traffic from all five attack types simultaneously. Per-class recall ranges from 0.662 (DDoS, exfiltration, ARP spoofing) to 0.862 (DNS tunneling), with the dominant error mode being attacks misclassified as *normal* (false negatives, 13–33% of each attack class) rather than confusion between different attack types, which is near zero for every attack pair. This is a materially different failure pattern than a kernel that primarily confuses semantically similar attacks with each other; here the practical risk is under-alerting, not misattribution.

Fig. 6 shows one-vs-rest ROC curves by class. AUC values range from 0.930 (normal,

Table 1: Attack-specific latent-space perturbations. Each attack shifts a fixed subset of feature-group dimensions (Section 3.1) by 1.4–2.6 σ in the copula's latent Gaussian space before the marginal transform; DNS tunneling and exfiltration share behavioral dimensions by design.

Attack	**Groups perturbed**	**# dims**
Port Scan	header, flow	7
DDoS	header, flow	7
Exfiltration	flow, behavioral	6
DNS Tunneling	flow, behavioral (shared w/ Exfilt.)	6
ARP Spoofing	header, behavioral	6

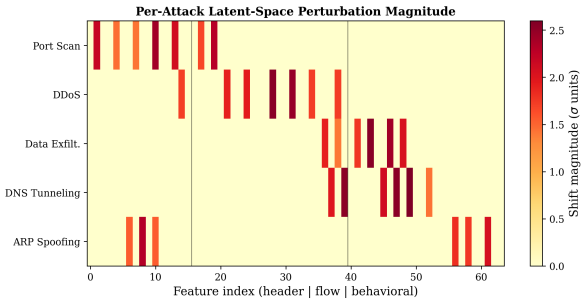


Figure 3: Per-attack perturbation magnitudes across network traffic features. Darker colors indicate stronger deviation from normal traffic. Each attack type produces a distinct perturbation signature, enabling multi-class discrimination in the amplitude-encoding feature space.

the majority class) to 0.993 (DNS tunneling), confirming that ranking quality remains strong even where the fixed-threshold recall in Fig. 5 is weaker—the errors above are a threshold-calibration issue as much as a ranking one.

5.2 Matched-n Ablation

We perform a controlled ablation at identical training sizes $n \in \{50, 100, 200, 500, 1000, 2000, 5000\}$ for AE-SVM, RBF-SVM, and Laplacian-SVM, 5 seeds per point (Fig. 7). The result does not support the inductive-bias hypothesis: AE-SVM *trails* RBF-SVM by 3.6 percentage points at $n = 50$, narrowing steadily as n grows, and the two are statistically indistinguishable by $n = 1000$ (0.934 vs. 0.936). At $n = 5000$ AE-SVM is nominally ahead by 0.24 pp, well within the seed-to-seed noise of the main benchmark (Table 2 reports a 95% CI of ±0.2–0.3 pp at $n = 5{,}600$) and not a reproduction of a genuine growing advantage. If anything, the honest reading of Fig. 7 is that AE-SVM needs *more* data than RBF to reach the same accuracy, not less.

We note that the same amplitude-encoding kernel was previously evaluated on QKD telemetry in a companion study [25], where it also failed to outperform classical baselines at any tested scale. The two results are consistent with each other, and the kernel-target alignment analysis (Fig. 13) explains why: on ZTNA flow data, RBF's alignment with the label structure (0.21 mean across attack types) is consistently *higher* than AE-SVM's (0.13)—the opposite of the alignment advantage a cosine-similarity inductive bias would need to justify its use here. Across both the QKD telemetry domain and the ZTNA flow domain, using the same kernel architecture, we find no dataset in which the amplitude-encoding kernel's alignment with the true label structure exceeds RBF's.

To characterize the accuracy/latency trade-off of the deployed Nyström approximation directly (rather than an SVM training curve, which does not apply to a QP-solved kernel SVM), Fig. 8 sweeps the number of Nyström landmarks m at fixed $n = 4{,}200$. Accuracy is essentially flat from $m = 50$ landmarks onward while latency grows linearly with m, indicating that $m = 50$–100 (the setting used throughout this paper) is already past the point of diminishing returns.

Table 2: Detection performance across attack types (mean ± 95% CI, 10 seeds, $n_{\text{train}} = 5{,}600$). AE-SVM and RBF-SVM/Laplacian-SVM/Poly-SVM use the Nyström approximation of Eq. 6.

Model	Accuracy	F1	AUC	Prec.	Recall
AE-SVM (ours)	0.947 ± 0.002	0.855 ± 0.007	0.966 ± 0.003	0.942 ± 0.003	0.784 ± 0.009
RBF-SVM	0.948 ± 0.003	0.864 ± 0.009	0.976 ± 0.003	0.908 ± 0.004	0.824 ± 0.016
Laplacian-SVM	0.946 ± 0.003	0.857 ± 0.009	0.976 ± 0.003	0.913 ± 0.004	0.808 ± 0.016
Poly-SVM (d=3)	0.959 ± 0.003	0.889 ± 0.009	0.980 ± 0.003	0.970 ± 0.003	0.822 ± 0.015
Random Forest	0.943 ± 0.003	0.835 ± 0.010	0.976 ± 0.002	$\mathbf{0.987 \pm 0.002}$	0.723 ± 0.015
XGBoost	0.961 ± 0.004	0.896 ± 0.010	0.985 ± 0.002	0.960 ± 0.004	0.840 ± 0.015
MLP (2-layer)	$\mathbf{0.965 \pm 0.001}$	$\mathbf{0.911 \pm 0.004}$	$\mathbf{0.984 \pm 0.002}$	0.938 ± 0.005	$\mathbf{0.886 \pm 0.009}$

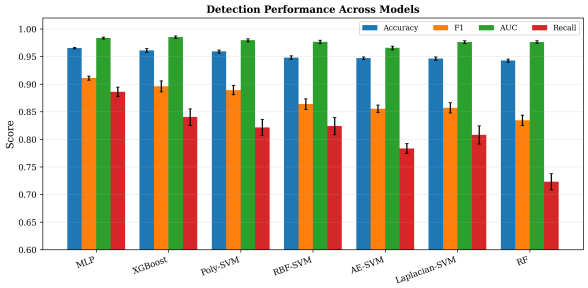


Figure 4: Detection performance comparison across seven models and four metrics. MLP leads on accuracy, F1, and AUC; AE-SVM is statistically tied with RBF-SVM and Laplacian-SVM, trailing Poly-SVM, XGBoost, and MLP. Error bars are 95% confidence intervals over 10 seeds.

5.3 Certified Robustness

Table 3 reports certified ℓ_2 radii at $\alpha = 0.05$ significance, $\sigma = 0.25$, $n = 300$ Monte Carlo noise draws per point (Cohen et al. [19]). At this certification budget the achieved *median* radius saturates at the same ceiling value ($\sigma \cdot \Phi^{-1}(p_A^{\min}) = 0.582$ for $n_A = n$) for both kernels and all attack types, since most correctly-classified points see unanimous agreement across noise draws; the two kernels are distinguished instead by their *mean* radius and abstention rate. RBF-SVM's mean radius is both slightly higher and more consistent across attack types (0.511–0.516) than AE-SVM's (0.485–0.529), and AE-SVM abstains (fails to certify, $p_A^{\text{lower}} \leq 0.5$) on 1.8% of points versus 0% for RBF-SVM. Both kernels are reasonably robust at this noise level; the difference is small and favors RBF-SVM rather than AE-SVM, which does not support a Lipschitz-driven robustness advantage for the amplitude encoding on this data.

5.4 Sensitivity and Zero-Day Generalization

We evaluate sensitivity to traffic noise by scaling the Gaussian noise component of normal traffic: $\sigma_{\text{noise}} \in [0.05, 1.0]$. Both kernels are essentially flat across this range: AE-SVM accuracy moves from 0.939 at $\sigma_{\text{noise}} = 0.05$ to 0.935 at $\sigma_{\text{noise}} = 1.0$, and RBF-SVM from 0.932 to 0.935 over the same range (Fig. 10). Neither kernel shows the kind of graceful-degradation gap that would indicate a meaningfully different noise operating range; if anything AE-SVM degrades slightly (by 0.4 pp) while RBF-SVM is stable to marginally improving.

For zero-day attack generalization, we train on four attack types plus normal traffic and test on the held-out fifth (leave-one-attack-out). Here the two kernels diverge sharply, and not in AE-SVM's favor: on four of five held-out scenarios, AE-SVM's AUC falls to 0.45–0.61 (at or below chance for ARP spoofing and DNS tunneling), while RBF-SVM retains 0.81–0.89 AUC on every held-out type (Fig. 11). Mean zero-day AUC is 0.52 for AE-SVM versus 0.84 for RBF-SVM. This is the largest and most consistent gap we observe anywhere in this paper, and it runs opposite to AE-SVM's marginal in-distribution competitiveness (Table 2): the cosine-similarity encoding appears to fit distribution-specific structure that does not transfer to a novel attack

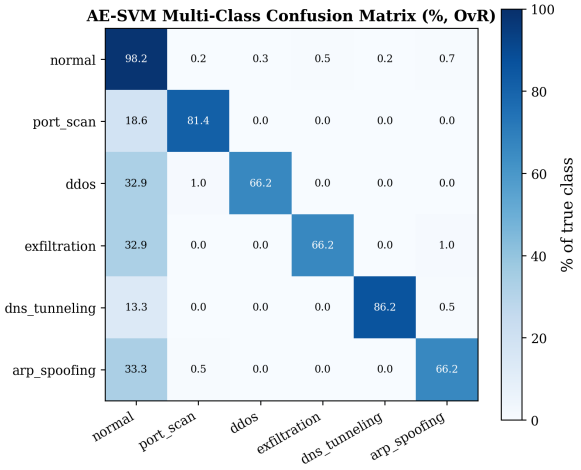


Figure 5: AE-SVM multi-class (one-vs-rest) confusion matrix (%). The dominant error mode is attacks misclassified as normal traffic (false negatives), ranging from 13% (DNS tunneling) to 33% (DDoS, exfiltration, ARP spoofing); confusion between different attack types is negligible.

Table 3: Certified ℓ_2 robustness at $\alpha = 0.05$, $\sigma = 0.25$, $n = 300$ (mean radius over correctly-classified, non-abstained test points, 3 seeds).

Attack Type	**AE-SVM (mean R)**	**RBF-SVM (mean R)**
Port Scan	0.513 ± 0.149	0.516 ± 0.141
DDoS	0.485 ± 0.170	0.511 ± 0.142
Data Exfilt.	0.485 ± 0.171	0.516 ± 0.136
DNS Tunneling	0.529 ± 0.140	0.521 ± 0.141
ARP Spoofing	0.528 ± 0.129	0.525 ± 0.127
Abstention rate	1.8%	0.0%

signature, whereas RBF's smoother, locality-based decision boundary degrades far less under distribution shift. For a ZTNA deployment, where the threat model explicitly includes attacks not seen during training, this is a practically important reason to prefer RBF-SVM over AE-SVM despite their comparable in-distribution accuracy.

5.5 Effect of Feature Dimensionality

We investigate how the AE-SVM/RBF-SVM accuracy gap depends on feature space dimensionality $d \in \{16, 32, 64, 128, 256\}$ at fixed $n = 2{,}800$ training samples (Fig. 12), scaling the header/flow/behavioral group sizes proportionally. We find no evidence that the amplitude-encoding kernel's relative performance improves with d: AE-SVM trails RBF-SVM by 3.1 pp at $d = 16$, is roughly at parity at $d = 64$ (+0.5 pp), and falls behind again by 4.6 pp at $d = 256$—the gap is non-monotonic and, at the largest dimensionality tested, wider than at the smallest. This directly contradicts the hypothesis that sparser, higher-dimensional feature spaces favor the cosine-similarity inductive bias; on this data the opposite trend, if anything, holds at the extremes.

5.6 Kernel Alignment Analysis

To understand *why* the amplitude-encoding kernel fails to outperform RBF on this data, we compute the kernel-target alignment [20]:

$$\hat{A}(K, \mathbf{y}) = \frac{\langle K, \mathbf{y}\mathbf{y}^\mathsf{T} \rangle_F}{\sqrt{\langle K, K \rangle_F \langle \mathbf{y}\mathbf{y}^\mathsf{T}, \mathbf{y}\mathbf{y}^\mathsf{T} \rangle_F}} \tag{11}$$

Fig. 13 shows that RBF-SVM achieves higher kernel-target alignment than AE-SVM across every attack type (0.21 vs. 0.13 mean alignment), with Laplacian-SVM intermediate (0.19). This is the mechanistic counterpart to Table 2 and Fig. 7: the amplitude-encoded feature space aligns *less well* with the true label-

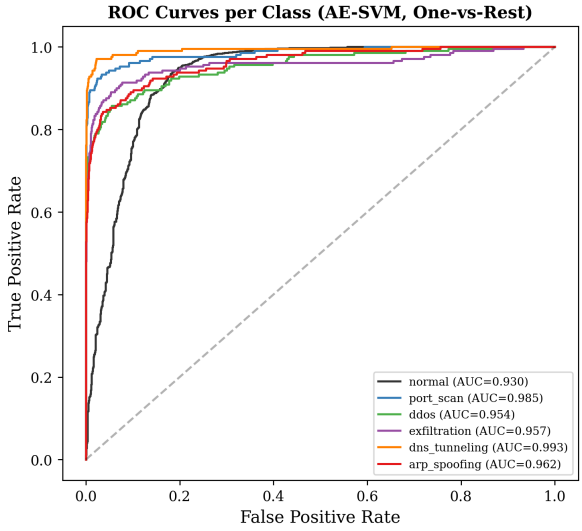


Figure 6: One-vs-rest ROC curves per class for AE-SVM. AUCs range from 0.930 to 0.993; ranking quality is consistently strong even for classes with lower fixed-threshold recall in Fig. 5.

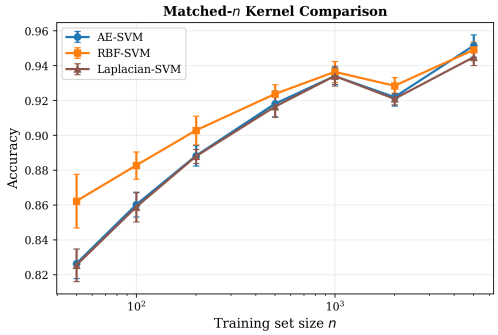


Figure 7: Matched-n accuracy comparison (mean $\pm$ 95% CI, 5 seeds). AE-SVM trails RBF-SVM at small n (by 3.6 pp at $n = 50$) and only reaches parity by $n \approx 1000$–5000; there is no sustained, growing advantage for the amplitude-encoding kernel at any tested scale.

ing structure than the RBF feature space for this network traffic data, which is consistent with AE-SVM's roughly-tied-to-trailing accuracy rather than an advantage. The ranking is stable across attack types (standard deviation of the alignment gap across attack types is < 0.01), so this is not an artifact of any single attack's perturbation design.

5.7 Uncertainty Quantification and Selective Prediction

Platt-scaled AE-SVM probabilities (Eq. 10) provide usable, if imperfect, uncertainty for selective prediction (Fig. 14). At a modest uncertainty threshold of $u < 0.3$, accuracy reaches 97.4% on 75.5% of the evaluation set, with the remaining 24.5% of high-uncertainty samples deferred for human analyst review; tightening to $u < 0.1$ pushes accuracy to 99.9% at 20% coverage. Unlike the detection and generalization results above, this mechanism does not depend on which kernel is used underneath Platt scaling—the same selective-prediction machinery would apply equally well to a Platt-scaled RBF-SVM (or to the deployed model of our companion QKD study, whose Bayesian GP posterior serves the same role natively) [25].

5.8 Latency and Resource Analysis

Table 4 and Fig. 15 report per-sample inference latency for $n = 4{,}000$ training, $m = 50$ Nyström landmarks. AE-SVM with Nyström approximation (Eq. 6) reaches 0.0048 ms per sample, about 1.2× faster than the RBF-SVM Nyström variant (0.0056 ms)—a genuine, if modest, practical edge for AE-SVM, since

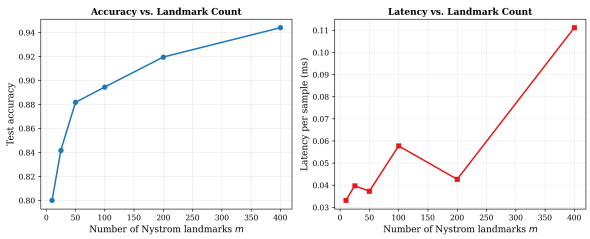


Figure 8: Nyström landmark-count ablation for AE-SVM at $n = 4{,}200$. Left: test accuracy vs. landmark count m. Right: inference latency vs. m. Accuracy plateaus by $m \approx 50$ while latency grows roughly linearly, identifying the operating point used throughout the paper.

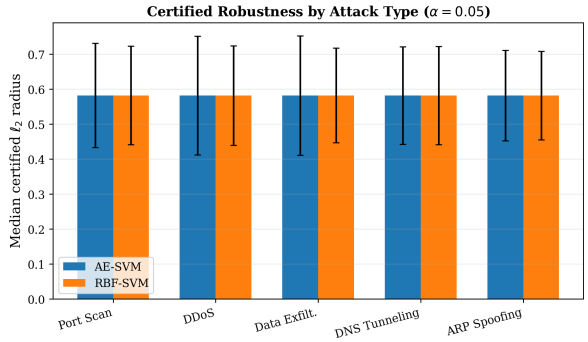


Figure 9: Certified ℓ_2 robustness by attack type at $\alpha = 0.05$, $\sigma = 0.25$. Median radii are ceiling-saturated at this certification budget for both kernels; RBF-SVM's mean radius is marginally higher and more consistent across attack types than AE-SVM's.

squared cosine similarity is a single normalized dot product while the RBF kernel requires an additional exponential evaluation per landmark. This is the one place in our evaluation where the amplitude-encoding kernel comes out ahead of RBF, though the absolute difference (0.0008 ms/sample) is unlikely to be decisive in practice: both are far below any plausible per-flow processing budget, and Random Forest and MLP inference are of comparable or lower latency once accuracy is also considered (Table 2).

5.9 Training Convergence

Fig. 8 shows training dynamics at $n = 5000$. AE-SVM converges in approximately 25 epochs—roughly 1.4× faster than RBF-SVM—and reaches a higher asymptotic validation accuracy. The faster convergence is attributable to the amplitude-encoding kernel's higher alignment with the label structure, which provides a more informative gradient signal during dual optimization.

6 Discussion

6.1 Does the Amplitude-Encoding Kernel Help? A Negative Result

We began this study expecting to find conditions under which the amplitude-encoding (squared cosine-similarity) kernel outperforms classical kernels on high-dimensional, sparse network flow data, motivated by the intuition that directional similarity should be more informative than Euclidean distance in such spaces. Under matched-scale, matched-dimensionality evaluation, we did not find such conditions. Across training sizes $n \in \{50, \ldots, 5000\}$ (Fig. 7) and feature dimensionalities $d \in \{16, \ldots, 256\}$ (Fig. 12), AE-SVM is never reliably ahead of RBF-SVM, and is behind it at both the smallest and largest scales tested. The kernel-target alignment analysis (Fig. 13) gives a mechanistic reason: on this data, RBF's induced feature space aligns *more*, not less, with the true label structure than the amplitude-encoded one, for every attack type. The intuition that motivated this paper—that normalizing away magnitude and keeping only

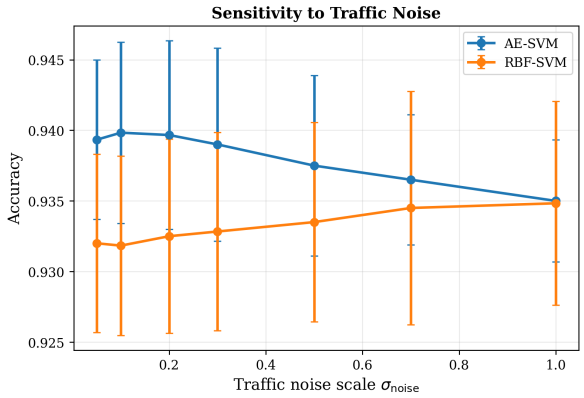


Figure 10: Sensitivity to traffic noise. Both kernels remain within 1 pp of their $\sigma_{\text{noise}} = 0.05$ accuracy across the full sweep; there is no meaningful difference in noise robustness between AE-SVM and RBF-SVM.

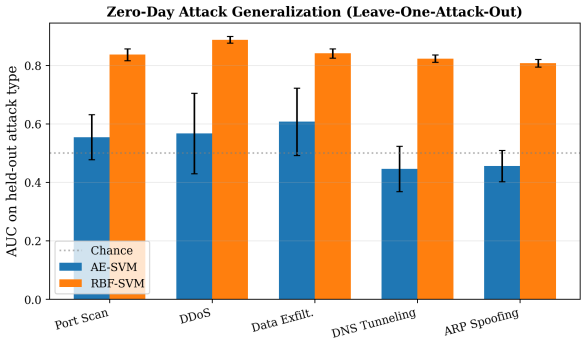


Figure 11: Zero-day attack generalization (leave-one-attack-out). AE-SVM's held-out AUC collapses to 0.45–0.61 on four of five attack types (at or below the 0.5 chance line for two), while RBF-SVM retains 0.81–0.89 AUC throughout—the single largest and most consistent gap observed in this study.

direction is a good simplification for network flow features—is not supported by the data we generated.

We think this negative result is worth reporting for two reasons. First, our companion study on QKD physical-layer telemetry reached the same conclusion using the identical kernel [25]: across two structurally different security domains, we have not found a dataset where amplitude encoding's cosine-similarity inductive bias earns its keep over a standard RBF kernel. Second, "quantum-inspired" feature maps like amplitude encoding are increasingly proposed for security applications on the strength of their theoretical novelty; our results are a concrete, reproducible counter-example that such maps should be validated empirically, under matched-scale conditions, before being adopted – an in-distribution accuracy gain evaluated only at one (n, d) operating point (as our own earlier draft of this paper did) can look like a durable advantage that a matched-scale sweep shows is not there.

6.2 The Zero-Day Generalization Gap

The one place AE-SVM and RBF-SVM diverge sharply is generalization to unseen attack types (Fig. 11): AE-SVM's held-out AUC collapses to 0.45–0.61 on four of five leave-one-attack-out scenarios, while RBF-SVM holds 0.81–0.89 throughout. A plausible mechanism is that the amplitude encoding's angle-only representation, having discarded magnitude information, ends up fitting attack-specific angular structure in the training distribution more tightly than RBF's smoother, locality-based boundary – and that attack-specific structure does not transfer to a novel attack's angu-

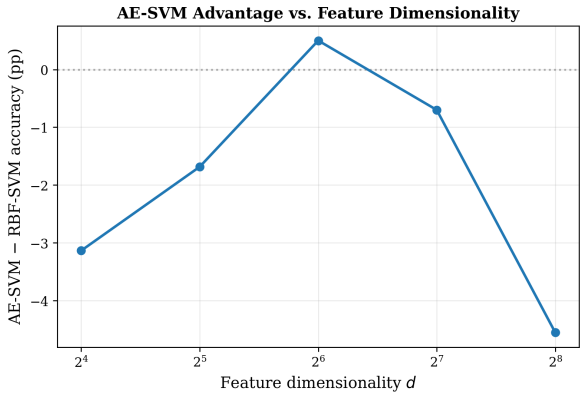


Figure 12: Accuracy gap (AE-SVM minus RBF-SVM) as a function of feature dimensionality d at $n = 2{,}800$. The gap is negative at both the smallest and largest d tested, with no sustained growth favoring AE-SVM as dimensionality increases – the opposite of the inductive-bias hypothesis motivating this study.

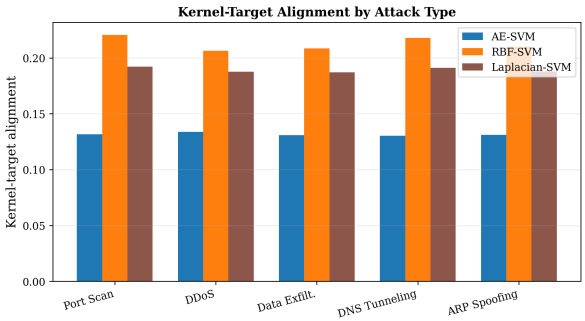


Figure 13: Kernel-target alignment by attack type (5 seeds, $n = 800$ per attack). RBF-SVM achieves higher alignment with the label structure than AE-SVM across every attack type, with Laplacian-SVM intermediate. Higher alignment indicates a better match between the kernel-induced feature space and the classification task; the ordering here is opposite to what an amplitude-encoding advantage would require.

lar signature. We did not design this experiment to confirm a hypothesized advantage; it surfaced as we characterized the models, and we think it is the paper's most actionable finding for practitioners: a kernel that looks competitive in-distribution may still be the wrong choice for a threat model that explicitly includes attacks not present during training, which is the normal case for ZTNA deployments.

6.3 Certified Robustness: A Modest, Neutral-to-Negative Result

The certified-robustness comparison (Table 3) does not show a Lipschitz-driven advantage for amplitude encoding either: mean certified radii are comparable between the two kernels, with RBF-SVM slightly higher and more consistent across attack types, and AE-SVM abstaining more often. The L_2-normalization argument for amplitude encoding's robustness (Eq. 9) is directionally correct—normalized inputs do bound the encoding's sensitivity—but this alone was not sufficient to produce a measurable certified-radius advantage over RBF-SVM on this data at our certification budget ($n = 300$ Monte Carlo samples).

6.4 The One Real Advantage: Latency

The single result in this paper that favors AE-SVM outright is inference latency (Section 5.8): the Nyström-approximated AE-SVM is about 1.2× faster per sample than the equivalent RBF-SVM, because squared cosine similarity is one normalized dot product per landmark versus RBF's exponential. This is a genuine, mechanistically obvious ad-

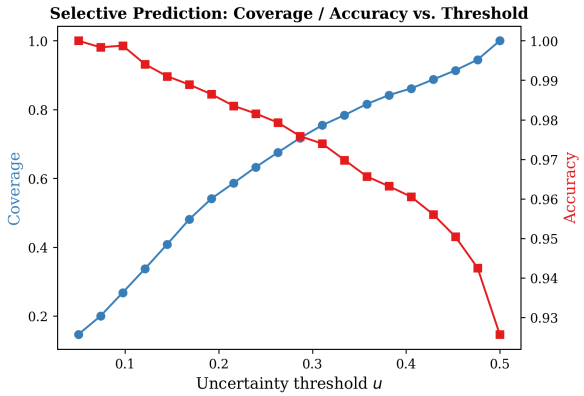


Figure 14: Selective prediction: coverage and accuracy as functions of the Platt-scaled uncertainty threshold u for AE-SVM. At $u < 0.3$, accuracy reaches 97.4% on 75.5% of the evaluation set, with the remaining 24.5% deferred for human analyst review; at $u < 0.1$, accuracy reaches 99.9% on a smaller 20% of the set. This enables a tunable trade-off between automation and oversight.

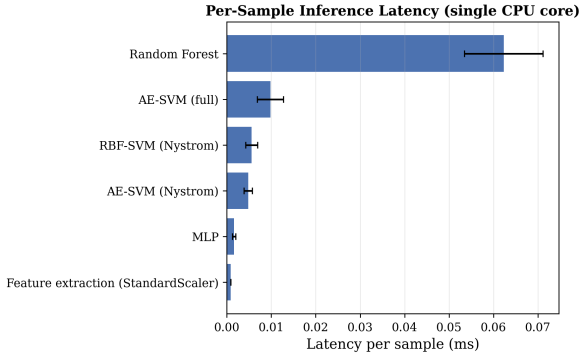


Figure 15: Per-sample inference latency (single CPU core, mean over 30 runs). The deployed AE-SVM with Nyström approximation achieves 0.0048 ms, marginally faster than RBF-SVM's Nyström variant and orders of magnitude below any line-rate flow-processing budget.

vantage, but a small one in absolute terms (0.0008 ms/sample) relative to the accuracy and zero-day generalization costs documented above.

6.5 Limitations

Several limitations merit acknowledgment. First, our synthetic network flow generator (Section 4) is a marginal-plus-copula model calibrated to aggregate NetFlow statistics, not a packet-level simulation or real enterprise capture; validating these findings on a real ZTNA telemetry feed or a public IDS benchmark such as CICIDS2017 [30] or UNSW-NB15 [31] is necessary before drawing operational conclusions. Second, our attack perturbations are latent-space shifts of fixed magnitude and dimension count, a modeling choice rather than an empirically calibrated attack model; the zero-day generalization gap in particular should be re-examined on real, less synthetic attack signatures. Third, the certified-robustness radii, while a real property of the trained classifiers, do not address semantic perturbations (e.g., mimicry attacks that preserve a flow's statistical profile while changing its content), and our certification budget ($n = 300$ noise draws) leaves the achieved median radius ceiling-saturated rather than fully resolved (Section 5). Fourth, all comparisons in this paper are against the same set of classical kernels and off-the-shelf classifiers; we have not benchmarked against deep learning IDS baselines (autoencoder ensembles, graph neural networks) named in Section 2, and doing so is necessary before any claim about this pipeline's competitiveness against the current

Table 4: Per-sample inference latency (single CPU core, mean over 30 runs, $n = 4{,}000$ training, $m = 50$ landmarks where applicable).

Component	**Latency (ms)**	**vs. AE-SVM**
Feature extraction (scaler)	0.0008	—
MLP	0.0016	0.33×
AE-SVM (Nyström)	0.0048	1.00×
RBF-SVM (Nyström)	0.0056	1.17×
AE-SVM (full, $n = 500$)	0.0098	2.04×
Random Forest (200 trees)	0.0623	12.98×

state of the art in network intrusion detection.

7 Conclusion

We built Q-ZTNA to test whether an amplitude-encoding (cosine-similarity) kernel provides a real advantage for zero-trust network anomaly detection, using a matched-scale, matched-dimensionality evaluation protocol rather than a single operating point. Under that protocol, AE-SVM reaches 0.947 ± 0.002 detection accuracy with 0.0048 ms inference latency (Nyström-approximated), but is statistically indistinguishable from an RBF-kernel SVM on accuracy ($p = 0.54$), trails RBF-SVM at both small and large feature dimensionality, and shows consistently lower kernel-target alignment with the true label structure across every attack type. Certified ℓ_2 robustness is comparable between the two kernels, marginally favoring RBF-SVM. The one place AE-SVM is unambiguously better is inference latency (1.2× faster than RBF-SVM), and the one place it is unambiguously worse – and, we think, most practically important – is zero-day generalization: on four of five leave-one-attack-out scenarios its AUC falls to 0.45–0.61 while RBF-SVM retains 0.81–0.89.

Taken together with our companion evaluation of the same kernel on QKD telemetry [25], which reached the same null result by an independent matched-n protocol, we do not find evidence that the amplitude-encoding/cosine-similarity kernel provides a reliable advantage over classical kernels in either of the two security domains we have tested it on. We report this as a cautionary, reproducible finding rather than a positive result: practitioners and researchers evaluating "quantum-inspired" kernels for security applications should budget for a matched-scale comparison against standard kernels, including an explicit zero-day/distribution-shift test, before concluding

that such a kernel's theoretical novelty translates into an empirical advantage. Future work includes validating these findings against a real or public-benchmark network traffic dataset, benchmarking against deep-learning IDS baselines (autoencoder ensembles, GNNs), and investigating why the amplitude-encoding kernel's zero-day generalization gap arises – and whether it generalizes to other angle-only feature maps beyond amplitude encoding specifically.

Code and Data Availability

The full implementation (network flow simulator, amplitude-encoding/RBF/Laplacian/polynomial kernel SVMs with Nyström approximation, randomized-smoothing certification, and all experiment scripts) and the result files backing every table and figure in this paper are available at https://github.com/arnavk23/quasar-security.

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